

Brainstorming & Idea Prioritization

Subscription Waste Detector

AI Specialist BFSI Program

Document Purpose

This document presents the complete brainstorming and idea prioritization process followed before selecting the final project. It explains the identified financial problem, alternative ideas that were explored, evaluation criteria, project selection rationale, expected benefits and future opportunities.

Executive Summary

Project Overview

Every successful software solution begins by identifying a meaningful real-world problem before proposing a technical implementation. Rather than developing technology for its own sake, the objective of the brainstorming phase was to identify a financial challenge that could be effectively addressed using Artificial Intelligence while remaining practical, scalable and relevant to the Banking, Financial Services and Insurance (BFSI) domain.

The increasing adoption of digital subscription services has transformed the way individuals consume entertainment, education, productivity software, cloud storage and AI-powered applications. While these services provide significant convenience, recurring subscription payments often continue unnoticed, resulting in unnecessary monthly expenses and reduced financial awareness.

After evaluating multiple project ideas using predefined selection criteria, the Subscription Waste Detector was chosen as the final solution. The application enables users to upload expense records, detect recurring subscription payments, visualize monthly subscription spending, monitor renewal dates and receive AI-powered cost-saving recommendations through an interactive dashboard.

Purpose of this Document

This document explains the structured brainstorming process that ultimately resulted in selecting the Subscription Waste Detector as the final project for the AI Specialist BFSI Program.

Brainstorming Objectives

The brainstorming session was conducted with clearly defined objectives to ensure that the selected project solved a meaningful financial problem while remaining achievable within the available development timeline.

Identify a Real Financial Problem

Focus on a practical problem experienced by everyday users instead of building a technology-driven solution without a clear purpose.

Meaningful AI Integration

Ensure that Artificial Intelligence contributes directly to solving the identified problem by providing useful insights rather than acting as a secondary feature.

Technical Feasibility

Select a project that could be successfully designed, developed and demonstrated using the available technology stack and project duration.

Future Scalability

Choose a solution capable of supporting future enhancements without requiring major architectural changes.

Problem Identification

Background

The rapid growth of subscription-based digital services has changed the way individuals access entertainment, education, productivity software, cloud storage, fitness applications and AI-powered tools. Instead of making one-time purchases, users increasingly subscribe to monthly or annual plans, resulting in multiple recurring payments throughout the year.

Although individual subscription costs appear relatively small, the combined monthly expenditure often becomes significant. Since payments are processed automatically, many users lose visibility into their recurring expenses and continue paying for services that are rarely or no longer used.

This lack of financial awareness highlighted the need for an intelligent solution capable of automatically identifying recurring subscription payments and presenting meaningful financial insights to users.

Existing Challenges

Several common financial challenges were identified during the brainstorming process.

Forgotten Subscription Renewals

Automatic renewal mechanisms continue charging users even after the service is no longer actively utilized, resulting in avoidable recurring expenses.

Limited Expense Visibility

Users generally know their overall monthly expenditure but rarely understand how much is specifically spent on recurring subscriptions.

Manual Transaction Analysis

Reviewing lengthy bank statements or expense records manually is time-consuming and often fails to reveal recurring payment patterns.

Lack of Personalized Insights

Traditional expense tracking applications record transactions but provide limited intelligence regarding unnecessary subscriptions or potential cost savings.

Key Observations

- Subscription-based digital services continue to increase across multiple industries.
- Users frequently underestimate the total amount spent on recurring subscriptions.
- Visualizing recurring payments improves financial awareness and spending decisions.
- Artificial Intelligence can identify recurring payment patterns more efficiently than manual analysis.
- Personalized recommendations can help users reduce unnecessary subscription expenses without affecting essential services.

Brainstormed Project Ideas

Multiple project concepts were explored before selecting the final solution. Each idea was evaluated based on practical usefulness, Artificial Intelligence integration, technical feasibility, scalability and expected user impact.

Project Idea	Description	AI Potential	User Value
Expense Tracker	Tracks daily income and expenditure while categorizing financial transactions.	Medium	Medium
Budget Planner	Helps users compare planned budgets with actual monthly expenses.	Medium	Medium
Bill Reminder System	Provides notifications for recurring utility and service bill payments.	Medium	Medium
Subscription Waste Detector	Automatically identifies recurring subscriptions and generates AI-powered financial recommendations.	High	High

Idea Evaluation Criteria

Each shortlisted project idea was evaluated using a structured set of criteria to ensure the final solution was practical, technically achievable, and aligned with the objectives of the AI Specialist BFSI Program.

Practical Relevance

The project should solve a genuine financial problem experienced by a large number of users in their daily lives.

Artificial Intelligence Integration

Artificial Intelligence should play a meaningful role in improving analysis, decision making, or user experience instead of being included merely as an additional feature.

Technical Feasibility

The complete solution should be implementable using the selected technology stack while remaining manageable within the available project duration.

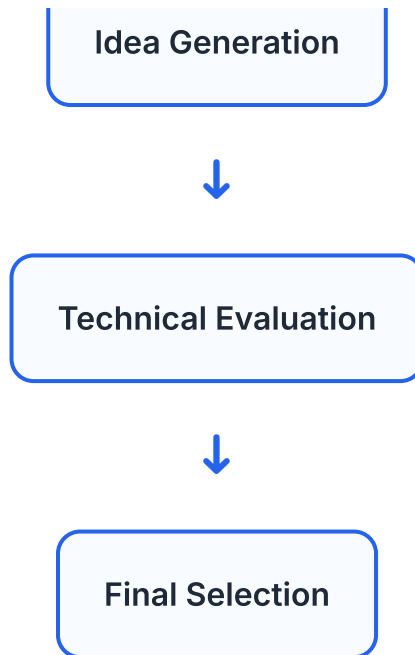
Future Scalability

The architecture should support future enhancements such as user authentication, persistent databases, notifications, and advanced analytics without requiring major redesign.

Project Selection Process

Problem Identification





Why Subscription Waste Detector Was Selected

Among all the evaluated concepts, the **Subscription Waste Detector** demonstrated the strongest balance between innovation, user value, technical feasibility, scalability, and Artificial Intelligence integration.

Selection Rationale

The selected solution addresses a common financial challenge faced by modern consumers who manage multiple subscription-based services. Instead of simply tracking expenses, the application analyzes uploaded transaction data to identify recurring subscription payments, provides meaningful visualizations of monthly subscription spending, reminds users about upcoming renewals, and generates AI-powered recommendations for reducing unnecessary recurring expenses.

The project also provides an excellent demonstration of practical Artificial Intelligence within the BFSI domain by transforming raw financial data into actionable insights that improve personal financial decision making.

"Effective financial decisions begin with complete visibility into recurring expenses. Artificial Intelligence transforms transaction data into meaningful insights that enable users to optimize subscription spending."

Subscription Waste Detector

Document 01

Technical Documentation

Expected Benefits

The Subscription Waste Detector delivers value to both end users and organizations by improving financial awareness and enabling intelligent subscription management.

Benefits for Users

- Improved visibility into recurring subscription expenses.
- Early identification of unnecessary or inactive subscriptions.
- Better financial planning through spending analysis.
- Personalized AI-powered cost optimization recommendations.
- Centralized monitoring of recurring payments.

Benefits for Organizations

- Demonstrates practical AI implementation in personal finance.
- Supports financial wellness initiatives.
- Provides opportunities for future fintech integrations.
- Encourages data-driven financial decision making.
- Creates a scalable foundation for future product enhancements.

Planned Functional Features

The first version of the Subscription Waste Detector focuses on delivering practical functionality while maintaining a simple and user-friendly experience.

Module	Description
Expense Upload	Upload CSV or Excel files containing transaction records for analysis.
Manual Expense Entry	Allow users to manually record individual financial transactions.
Recurring Subscription Detection	Automatically identify recurring payments based on transaction frequency and merchant information.
Subscription Dashboard	Display detected subscriptions, monthly recurring costs and overall spending summaries.
AI Recommendations	Generate personalized suggestions for reducing unnecessary subscription expenses using Groq AI.
Renewal Reminder	Notify users about upcoming subscription renewal dates to improve financial awareness.

Future Enhancements

The current implementation establishes a strong foundation that can be extended with additional intelligent capabilities in future versions.

Secure User Authentication

Enable multiple users to securely access their own subscription data and financial insights.

Persistent Database

Store expense history and subscription information using a permanent database instead of temporary session data.

Bank Statement Integration

Import transaction records directly from supported bank statements for faster expense analysis.

Advanced Financial Analytics

Provide long-term spending trends, predictive subscription analysis and detailed financial reports.

Final Decision

Following the brainstorming, analysis and evaluation process, the Subscription Waste Detector was selected as the final project because it effectively combines Artificial Intelligence with practical financial analytics while addressing a common real-world problem experienced by modern digital consumers.

Conclusion

Summary

The brainstorming and idea prioritization phase established a clear direction for the successful development of the Subscription Waste Detector. Rather than beginning with technology, the project started by identifying a meaningful financial problem and evaluating multiple possible solutions before selecting the most practical alternative.

The selected solution demonstrates how Artificial Intelligence can improve personal financial management by identifying recurring subscription payments, visualizing spending behaviour and generating actionable recommendations for reducing unnecessary expenses. This structured approach provided a strong foundation for the subsequent phases of requirements analysis, system design, implementation, testing and deployment.

Document Outcome

This document successfully captures the complete brainstorming and project selection process that led to the development of the Subscription Waste Detector for the AI Specialist BFSI Program.